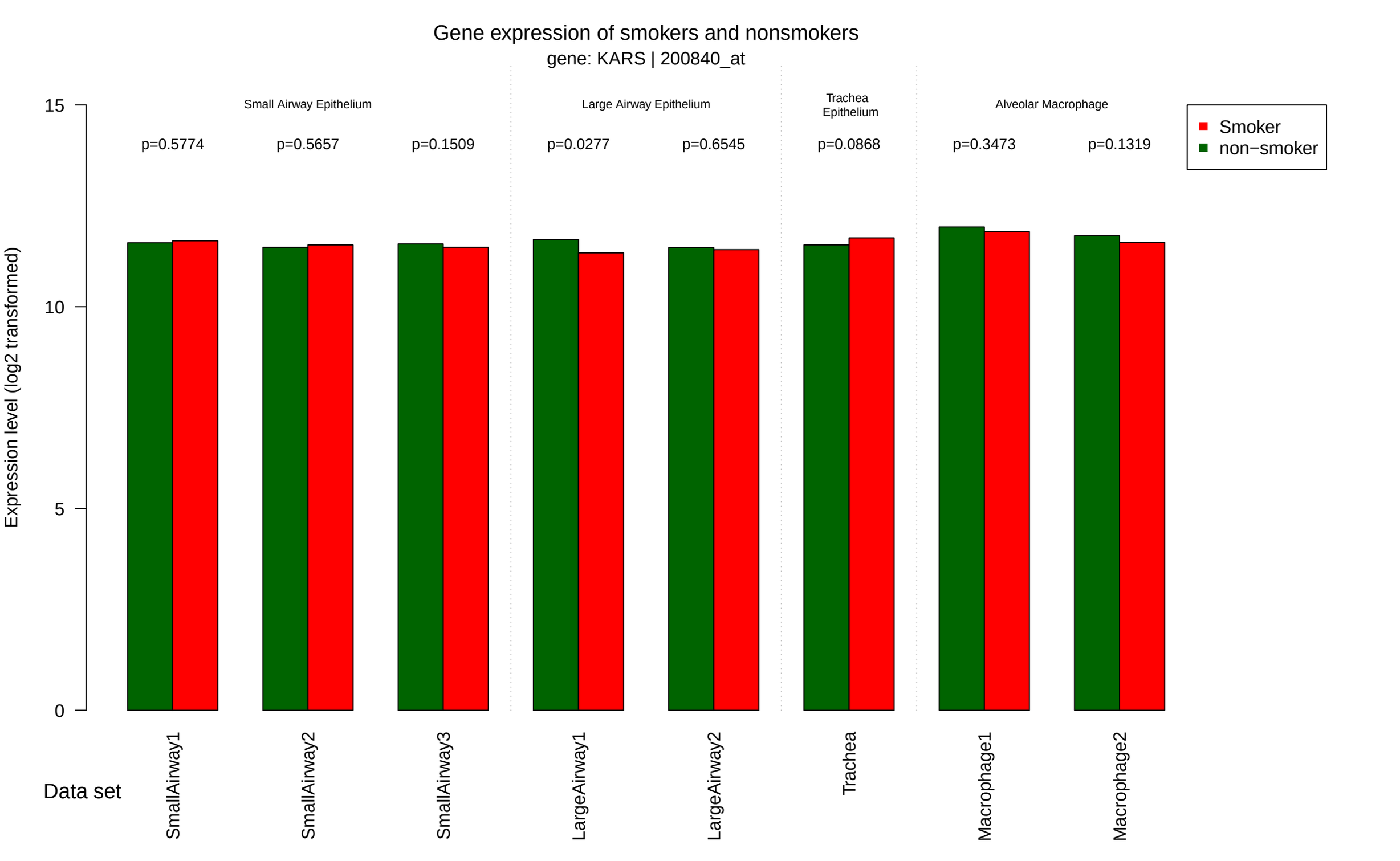
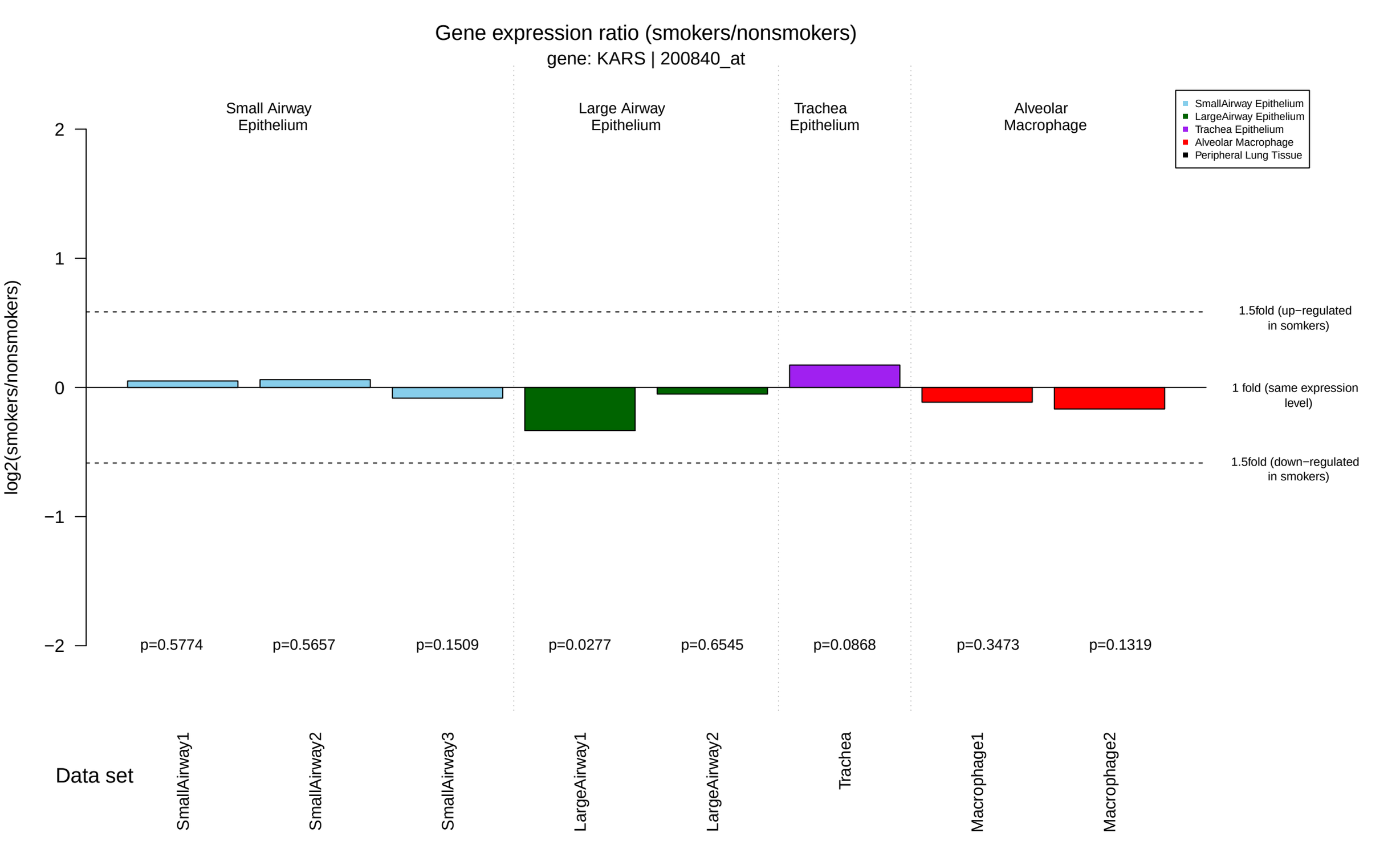
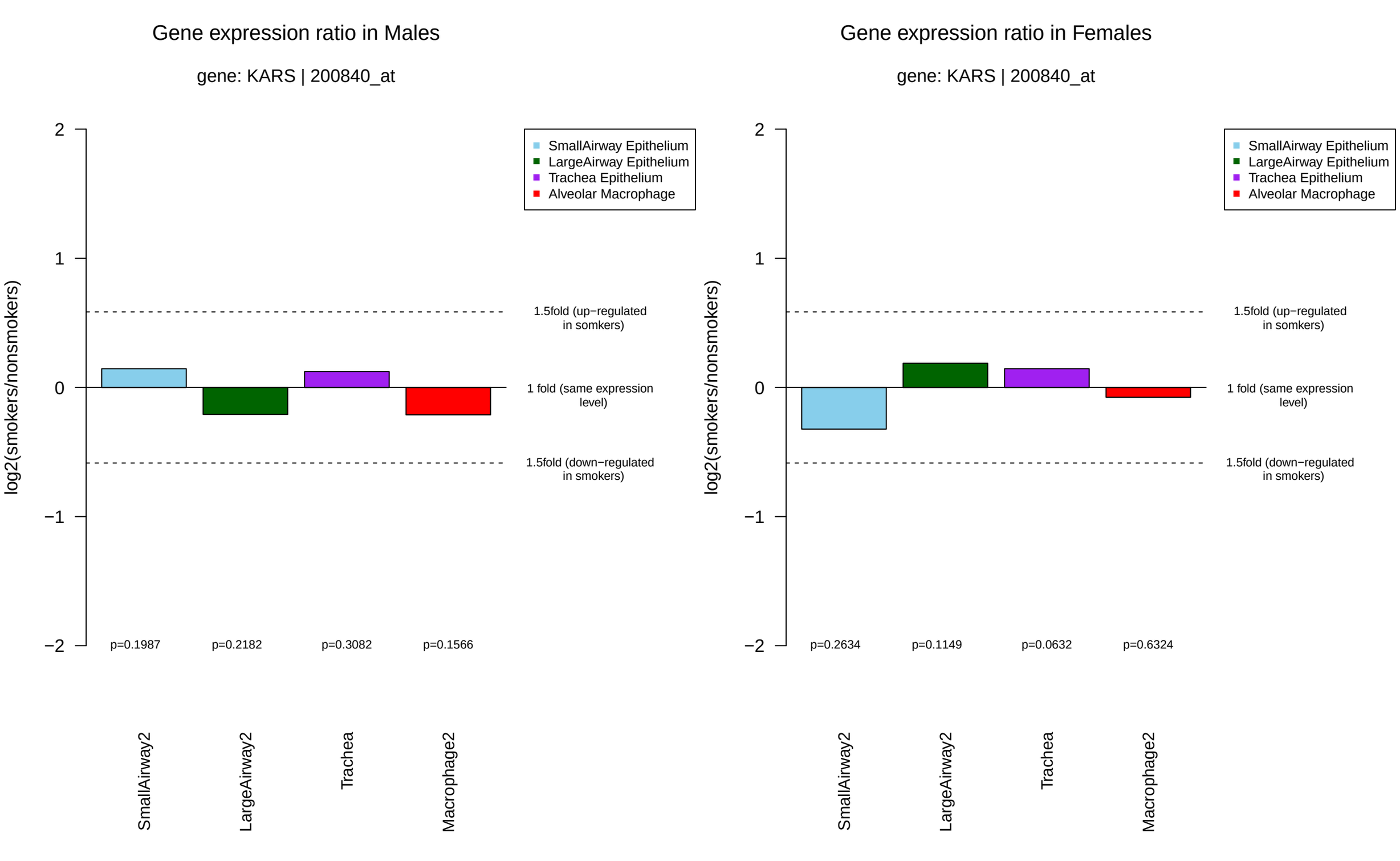
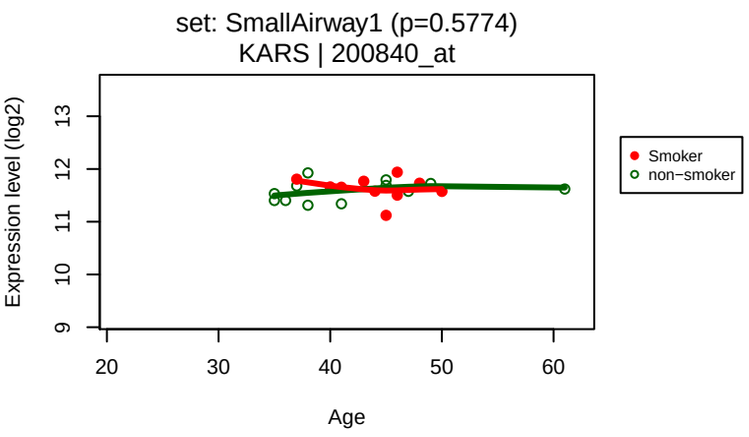
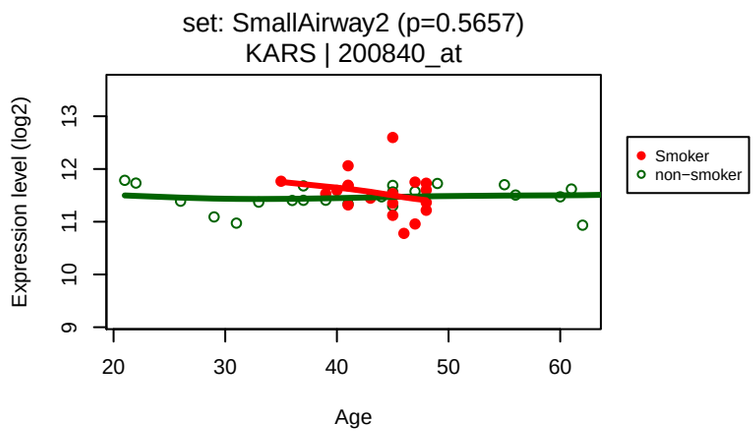
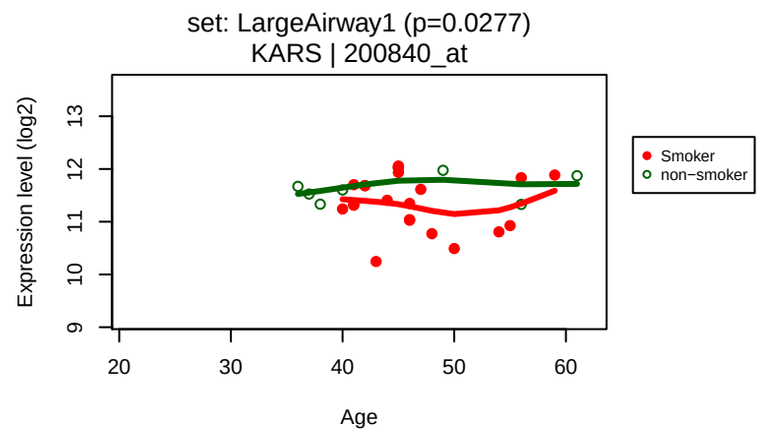

Gene expression of smokers and nonsmokers
gene: KARS | 200840_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5774
p=0.5657
p=0.1509
p=0.0277
p=0.6545
p=0.0868
p=0.3473
p=0.1319
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

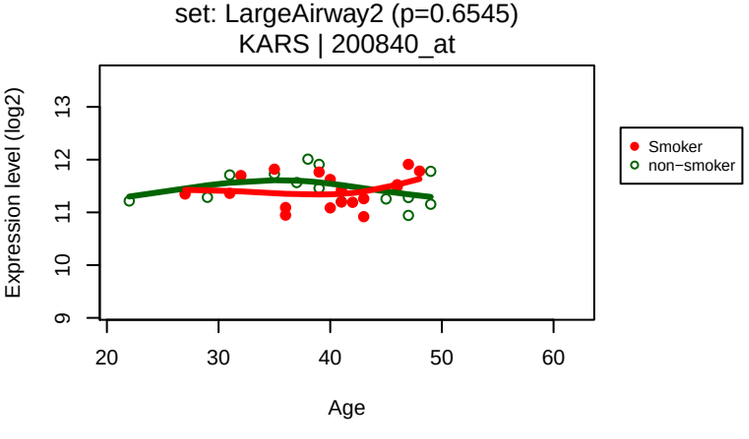

Gene expression ratio (smokers/nonsmokers)
gene: KARS | 200840_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5774
p=0.5657
p=0.1509
p=0.0277
p=0.6545
p=0.0868
p=0.3473
p=0.1319
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

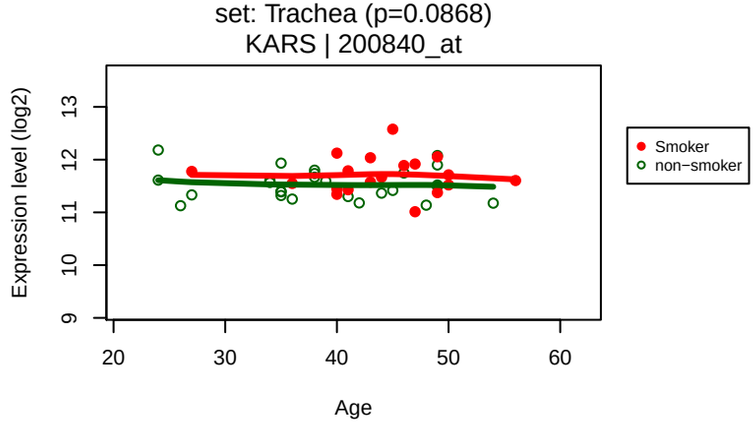

Gene expression ratio in Males
gene: KARS | 200840_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1987
p=0.2182
p=0.3082
p=0.1566
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KARS | 200840_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2634
p=0.1149
p=0.0632
p=0.6324
SmallAirway2
LargeAirway2
Trachea
Macrophage2

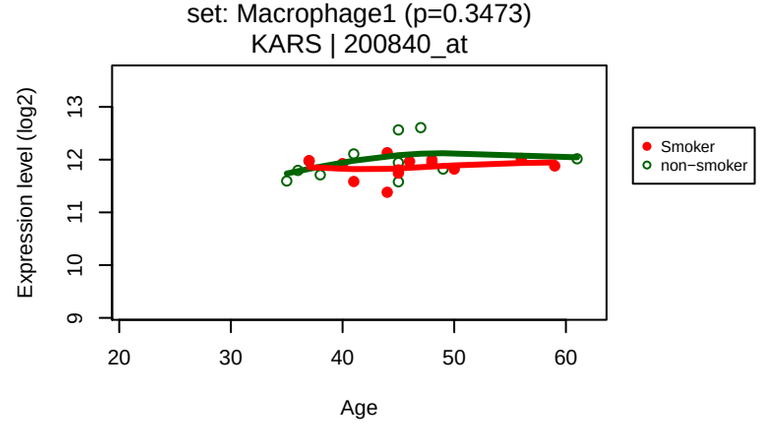

set: SmallAirway1 (p=0.5774)
KARS | 200840_at
Expression level (log2)
Age
Smoker
non−smoker

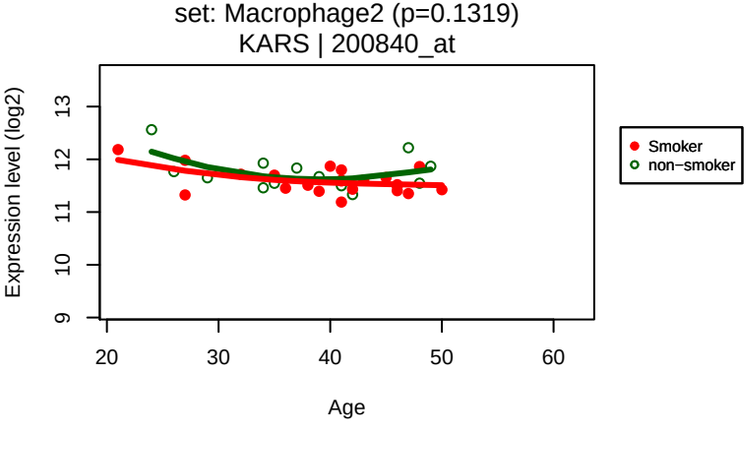

set: SmallAirway2 (p=0.5657)
KARS | 200840_at
Expression level (log2)
Age
Smoker
non−smoker

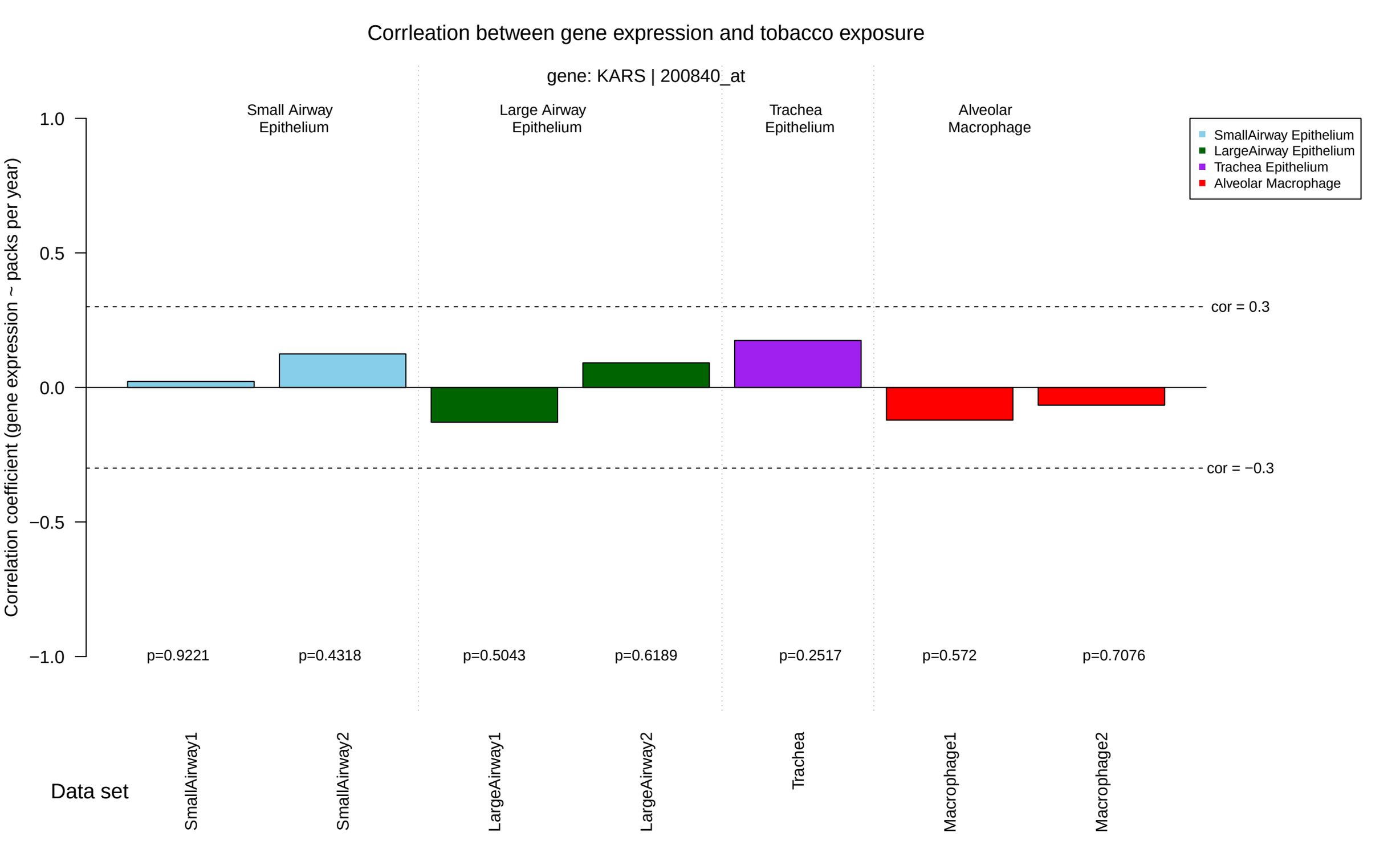

set: LargeAirway1 (p=0.0277)
KARS | 200840_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6545)
KARS | 200840_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0868)
KARS | 200840_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3473)
KARS | 200840_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1319)
KARS | 200840_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KARS | 200840_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9221
p=0.4318
p=0.5043
p=0.6189
p=0.2517
p=0.572
p=0.7076
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2